

**23CSE111 OBJECT ORIENTED PROGRAMMING**  
**S2 B.Tech CSE**

**Department of Computer Science and Engineering**  
**Amrita School of Computing, Amritapuri Campus**

**CASE STUDY**

**PHASE – I (Analysis)**

Problem Identification & Class Mapping (CO1)

Group No : D15

Team Members :

| Roll No          | Student Name        |
|------------------|---------------------|
| AM.SC.U4CSE25352 | Sreyak S            |
| AM.SC.U4CSE25314 | Dhananjayan Sreeraj |
| AM.SC.U4CSE25320 | Hari Sankar V       |
| AM.SC.U4CSE25354 | Tharak Gopal        |

1. Problem Title : **Airline Management System**

2. Problem Description :

The **Airline Management System** is designed to manage airline operations such as flight scheduling, passenger bookings. Traditional manual systems often lead to overbooking, schedule conflicts, and poor customer experience.

This system digitizes airline operations by allowing passengers to search for flights, book tickets, and cancel reservations. Administrators can manage flights, view scheduled flights, and update airline data. The system ensures accurate seat availability, secure user data handling, and efficient reservation management.

### 3. Identified Entities (Classes)

| Class Name      | Description                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------|
| User            | Handles login and user credentials.                                                          |
| Admin           | Represents airline staff responsible for managing flights, schedules, and system operations. |
| Flight          | Represents a specific flight operated by the airline.                                        |
| Main            | Entry point of the system.                                                                   |
| Airport         | Represents airports and the gates used for boarding.                                         |
| Ticket          | Represents the travel document issued after a booking.                                       |
| GetFare         | Calculates ticket fare.                                                                      |
| Payment         | Handles payment transactions for bookings.                                                   |
| payByCard       | Interface for card payments.                                                                 |
| payByUPI        | Interface for UPI payments.                                                                  |
| payByNetBanking | Interface for net banking.                                                                   |
| FlightCSVReader | Reads flight data from CSV.                                                                  |
| Passenger       | Stores passenger information.                                                                |
| PassCheck       | Validates passenger/ user data.                                                              |
| PremiumTicket   | Represents the travel document issued after a booking premium.                               |

#### 4. Class Attributes and Behaviours

| Class Name    | Attributes                                                                                                                                                                                                                                                                | Methods (behaviours)                                                                                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Passenger     | -fname: String<br>- lname: String<br>-nationality: String<br>- phone: String<br>- email: String<br>- passport: String<br>- age: int<br>- dob: String<br>- seat: String                                                                                                    | Passenger(String,String,String,String,String,int,String,String                                          |
| Admin         |                                                                                                                                                                                                                                                                           | + authenticateAdmin(): boolean                                                                          |
| Flight        | - flightID: String<br>- aircraftID: String<br>- dateofdeparture: String<br>- dateofarrival: String<br>- timeofdeparture: String<br>- timeofarrival: String<br>- origin: Airport<br>- destination: Airport<br>- fare: double<br>- totalseats: int<br>- availableseats: int | + Flight(String,String,String,String,String,String,Airport,Airport,int,double)<br>+ create Flight: void |
| Airport       | - airportCode: String<br>- city: String<br>- country: String<br>- gate: String                                                                                                                                                                                            | + Airport(String,String,String,String)<br>+ getairportCode(): void<br>+ getcity(): void                 |
| PremiumTicket | - extraCharge: double                                                                                                                                                                                                                                                     | + PremiumTicket(Passenger,Flight,String,String,String,Payment,double)                                   |
| Ticket        | - passenger: Passenger<br>- PNR: String<br>- ticketNumber: String<br>- ticketStatus: String<br>- flight: Flight<br>- payment: Payment                                                                                                                                     | + Ticket(Passenger,Flight,String,String,String,Payment)<br>+ saveTicketToFile(): void                   |
| GetFare       |                                                                                                                                                                                                                                                                           | + getfare(String): double                                                                               |
| Payment       | - paymentID: String<br>- transactionID: String                                                                                                                                                                                                                            | + Payment(double)<br>+ payByCard(String,String,String,int): void                                        |

|                 |                                                                                            |                                                                                                                               |
|-----------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|                 | -sourceID: String<br>-amount: double<br>- paymentMethod: String<br>- paymentStatus: String | + payByUPI(String): void<br>+ payByNetBanking(String,String,String): void<br>+ getPaymentStatus(): String<br>+ process(): int |
| payByCard       |                                                                                            | +paybycard(String,String,String,int): void                                                                                    |
| payByNetBanking |                                                                                            | + paybynetbanking(String,String,String): void                                                                                 |
| payByUPI        |                                                                                            | + paybyupi(String):void                                                                                                       |
| User            |                                                                                            | + createAccount(): boolean<br>+authenticatorUser: boolean<br>+ showFlight(): void<br>+ bookFlight(): void<br>+ flow(): void   |
| FlightCSVReader |                                                                                            | - findFlightByID(String flightID): String                                                                                     |
| PassCheck       |                                                                                            | - passvalid(String): boolean                                                                                                  |
| Main            |                                                                                            | - <u>main(): void</u>                                                                                                         |